Multiple Schedulers in Kubernetes

Since you're studying **KCNA**, let's understand this in a beginner-friendly way.

1. First: What is a Scheduler?

The **Kubernetes Scheduler** decides **which Node should run a Pod**.

For example:

```
Pod
 |
 ▼
Scheduler
 |
  ├── Node 1 ❌ Busy
  ├── Node 2 ✅ Suitable
 └── Node 3 ❌ Not suitable
        |
        ▼
      Node 2
```

So remember:

📍 **Scheduler = Decides where a Pod should run.**

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2. What are Multiple Schedulers?

Normally, Kubernetes has **one default scheduler**:

kube-scheduler

But Kubernetes can also have **more than one scheduler**.

Multiple schedulers means running multiple scheduler processes in the same Kubernetes cluster, where different Pods can be assigned to different schedulers.

For example:

```
Kubernetes Cluster
      |
      ├── Scheduler 1
      |     └── General applications
      |
      └── Scheduler 2
            └── Special applications
```

---

3. Why Use Multiple Schedulers?

Sometimes the default scheduler's rules are not enough for a particular application.

You might want a special scheduler for a particular type of workload.

For example:

Scheduler 1

↓

Normal applications

Scheduler 2

↓

Special applications

This allows you to have **different scheduling policies** for different workloads.

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4. Real-Life Example 🏫

Imagine a school has two people assigning students to classrooms.

Scheduler 1 👨‍🏫

Handles normal students:

Students → Normal classrooms

Scheduler 2 👩‍🏫

Handles special students:

Special students → Special classrooms

Both are doing the same general job:

Deciding where students should go.

But they use different rules.

That's similar to **multiple schedulers**.

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5. How Does a Pod Choose a Scheduler?

A Pod can specify which scheduler should handle it using:

```
spec:
  schedulerName: my-scheduler
```

For example:

```
apiVersion: v1
kind: Pod

metadata:
  name: my-pod

spec:
  schedulerName: my-scheduler

  containers:
  - name: nginx
    image: nginx
```

Here:

schedulerName: my-scheduler

means:

"I want my-scheduler to schedule me."

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6. Default Scheduler

If you don't specify a scheduler:

spec:

  containers:

Kubernetes normally uses:

default-scheduler

So:

No schedulerName

↓

Default Scheduler

But:

schedulerName: my-scheduler

means:

Pod

↓

my-scheduler

---

7. Example with Two Schedulers

Imagine we have:

Scheduler 1

default-scheduler

Scheduler 2

gpu-scheduler

Normal Pod

spec:

  schedulerName: default-scheduler

Goes to:

default-scheduler

GPU Pod

spec:

  schedulerName: gpu-scheduler

Goes to:

gpu-scheduler

---

8. Important Point ⭐

Multiple schedulers **do not mean that all schedulers schedule every Pod**.

Instead, a Pod is associated with a particular scheduler through:

schedulerName:

The scheduler whose name matches the Pod's schedulerName handles that Pod.

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9. Visual Understanding

```
        Kubernetes Cluster
               |
     ┌──────────┴──────────┐
     |          |
     ▼          ▼
  default-scheduler   gpu-scheduler
```

|　　　　|

▼　　　　▼

Normal Pods　　GPU Pods

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10. Why Is This Useful?

Multiple schedulers can be useful when you need:

- Different scheduling policies
- Special workloads
- Custom scheduling logic
- Different rules for different applications

For example:

Normal applications

↓

Default Scheduler

GPU/AI applications

↓

Custom Scheduler

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11. Multiple Schedulers vs Node Affinity

Don't confuse them.

Node Affinity

You tell Kubernetes:

"I prefer or require a Node with these labels."

Pod

↓

Scheduler

↓

Node Affinity rules

↓

Suitable Node

Multiple Schedulers

You tell Kubernetes:

"Use this particular scheduler to decide where I should run."

Pod

↓

schedulerName

↓

Specific Scheduler

↓

Suitable Node

KCNA Quick Revision ⭐

Scheduler

📍 Decides **where Pods should run**.

Multiple Schedulers

📍 Allows **more than one scheduler** to operate in a Kubernetes cluster.

schedulerName

📍 Specifies **which scheduler should schedule a Pod**.

Example:

spec:

  schedulerName: my-scheduler

Default

If no scheduler is specified:

default-scheduler

is normally used.

🧠 **Easy Memory Trick**

ONE Scheduler

↓

Most Pods

↓

default-scheduler

MULTIPLE Schedulers

↓

Different Pods

↓

Different scheduling policies

One-line KCNA definition:

Multiple schedulers allow a Kubernetes cluster to use different scheduling components so that different Pods can be scheduled according to different policies.